

--- **1 DVT Deep Vein Thrombosis** • **Definition** Formation of a blood clot thrombus in a deep vein usually of the lower limb

- **Signs** Unilateral swelling calf pain/tenderness warmth and redness
- **Complication** Pulmonary embolism
- **Prevention** Early mobilization ankle pump exercises compression devices and anticoagulants when prescribed

--- **2 Scar Types** • **Definition** A scar is fibrous connective tissue formed during healing after tissue injury • **Types** • Normal scar Flat soft and within wound margins • Hypertrophic scar Raised but remains within wound margins • Keloid Raised scar extending beyond original wound margins • Atrophic scar Thin depressed scar

--- **3 Types of Antibiotics** • Cell wall inhibitors Penicillins cephalosporins • Protein synthesis inhibitors Aminoglycosides macrolides tetracyclines • Nucleic acid inhibitors Fluoroquinolones rifampicin • Antimetabolites Sulfonamides trimethoprim • Cell membrane disruptors Polymyxins

--- **4 Dehydration Manifestations** • **Definition** Dehydration is a condition in which body water loss exceeds water intake • **Manifestations** • Thirst • Dry mouth and mucous membranes • Reduced/dark urine • Weakness and dizziness • Tachycardia • Hypotension in severe cases

--- **5 Surgical Drains** • **Definition** Surgical drains are tubes/devices used to remove blood pus serous fluid or other unwanted collections from a wound or body cavity • **Examples** • Intercostal drain ICD • Hemovac • Jackson Pratt drain • T tube • **Purpose** Prevent fluid accumulation and facilitate healing

--- **6 Types of Anaesthesia** • Local anaesthesia Loss of sensation in a limited area without loss of consciousness • Regional anaesthesia Blocks sensation in a larger region • Example spinal epidural • General anaesthesia Drug induced reversible unconsciousness with loss of protective reflexes and sensation • Sedation Reduced anxiety/awareness with varying levels of consciousness

--- **7 Types of Wounds** • Open wound Break in skin continuity • Example laceration abrasion • Closed wound Skin remains intact but underlying tissue is damaged • Example contusion • Acute wound Usually heals in an expected time • Example surgical incision • Chronic/non healing wound Fails to progress normally through healing • Example diabetic or pressure ulcer • Surgical wounds may also be classified as clean clean contaminated contaminated and dirty/infected

--- **8 Wound Healing** • **Definition** Wound healing is the biological process by which the body repairs damaged tissue and restores tissue continuity • **Phases** 1 Hemostasis stops bleeding 2 Inflammation removes bacteria and debris 3 Proliferation granulation tissue collagen angiogenesis and epithelialization 4 Remodeling collagen reorganization and scar maturation

-- **9 Gangrene** • **Definition** Gangrene is tissue necrosis caused by inadequate blood supply infection or both • **Types** • Dry gangrene Usually due to ischemia • Wet gangrene Necrotic tissue with infection • Gas gangrene Rapidly progressive infection with gas formation

--- **10 Cellulitis** • **Definition** Cellulitis is an acute bacterial infection of the skin and deeper subcutaneous tissues • **Clinical features** • Redness • Warmth • Swelling • Pain/tenderness • Fever may occur

- **Treatment** Antibiotics elevation and treatment of the underlying cause

--- **11 Abscess** • **Definition** An abscess is a localized collection of pus within a tissue or body cavity usually caused by infection • **Clinical features** • Pain • Swelling • Redness • Warmth • Fluctuant/pus filled swelling • Fever may occur • **Treatment** Drainage when indicated appropriate antibiotics

--- **12 Amputation** • **Definition** Amputation is the surgical removal of part or all of a limb • **Indications** Severe trauma gangrene peripheral vascular disease uncontrolled infection and malignancy • **Types** Above knee below knee above elbow and below elbow amputations • **Rehabilitation** Stump care strengthening ROM prosthetic training and gait training

-- **13 Phantom Limb** • **Definition** Phantom limb sensation is the feeling that an amputated limb is still present • **Phantom limb pain** Pain perceived in the missing limb • **Features** Burning tingling shooting pain or sensation of movement

- **Management** Desensitization mirror therapy graded motor imagery appropriate prosthetic rehabilitation and medical pain management

--- **14 Varicose Veins** • **Definition** Varicose veins are dilated elongated and tortuous superficial veins usually of the lower limbs caused mainly by venous valve incompetence • **Features** Visible twisted veins aching heaviness ankle edema and skin changes • **Management** Exercise leg elevation compression when appropriate and selected surgical/endovenous treatment

--- **15 Craniotomy** • **Definition** Craniotomy is a neurosurgical procedure in which a portion of the skull is temporarily removed to provide access to the brain and is usually replaced after surgery • **Indications** Brain tumour aneurysm hematoma vascular lesions and other intracranial conditions • **Difference** In craniectomy the bone flap is not immediately replaced

--- **16 Laminectomy** • **Definition** Laminectomy is a spinal operation involving removal of part or all of the vertebral lamina to decompress the spinal cord or nerve roots • **Indications** Spinal stenosis nerve root compression selected disc disease and spinal tumours • **Complications** Dural tear/CSF leak infection bleeding neurological injury and spinal instability

--- **18 Postoperative Complications** • **Definition** Postoperative complications are unwanted conditions occurring after surgery that may delay recovery or cause morbidity • **Common complications** • Shock • DVT • Wound infection • Atelectasis • Pneumonia • UTI • Pressure injuries • Hemorrhage • Wound dehiscence • Ileus

- **Physiotherapy prevention** Early mobilization breathing exercises circulation exercises ROM positioning and functional rehabilitation

////////// **Craniotomy Overview Indications and Complications** --- **Definition** A craniotomy is a neurosurgical procedure in which a portion of the skull is temporarily removed as a bone flap to provide access to the brain for diagnosis or treatment the bone flap is usually replaced at the end of surgery

--- **1 Overview of Craniotomy** A craniotomy provides the surgeon with access to intracranial structures while attempting to minimize damage to normal brain tissue The location and size of the craniotomy depend on the pathology being treated **Common types according to**

location • Frontal craniotomy • Temporal craniotomy • Parietal craniotomy • Occipital craniotomy • Pterional craniotomy • Retrosigmoid/suboccipital craniotomy It may be performed under general anesthesia while selected procedures involving areas responsible for speech or movement may be performed as an awake craniotomy to allow neurological monitoring during surgery **Basic procedure** 1 Patient is positioned and anesthetized 2 A scalp incision is made 3 Burr holes are created 4 A section of skull is removed as a bone flap 5 The dura is opened to expose the brain 6 The required intracranial procedure is performed 7 The dura is closed 8 The bone flap is replaced and fixed 9 The scalp is closed --- **2 Indications of Craniotomy** Craniotomy may be performed for diagnostic therapeutic or decompressive purposes **A Brain tumours** • Meningioma • Glioma • Metastatic brain tumour • Pituitary and parasellar lesions • Other intracranial tumours **B Cerebrovascular conditions** • Cerebral aneurysm • Arteriovenous malformation AVM • Intracerebral hemorrhage • Selected vascular malformations **C Traumatic brain injury** • Epidural hematoma • Acute subdural hematoma • Depressed skull fracture • Intracranial foreign body **D Raised intracranial pressure / brain swelling** • Selected cases of severe cerebral edema or swelling following trauma or stroke **E Infectious lesions** • Brain abscess • Subdural empyema **F Other indications** • Epilepsy surgery to remove a seizure focus • Microvascular decompression for selected cranial nerve disorders • Biopsy or removal of intracranial lesions • Selected functional neurosurgical procedures --- **3 Complications of Craniotomy** Complications can be neurological surgical systemic or wound related **A Neurological complications** • Brain injury • New neurological deficit • Limb weakness or paralysis • Sensory impairment • Speech or language disturbance • Visual disturbance • Problems with balance and coordination • Cognitive or memory changes • Seizures • Stroke • Cerebral edema **B Hemorrhagic complications** • Intracranial hemorrhage • Epidural or subdural hematoma • Excessive blood loss • Venous sinus injury **C CSF related complications** • Cerebrospinal fluid CSF leak • Hydrocephalus • Pneumocephalus **D Infectious complications** • Surgical site infection • Meningitis • Bone flap infection • Brain abscess or empyema **E Other complications** • Headache • Nausea and vomiting • Scalp numbness or pain • Cosmetic deformity • Temporalis muscle weakness/atrophy • Deep vein thrombosis and other complications related to immobility --- **4 Physiotherapy Implications** Although the question focuses on craniotomy this section is useful for a physiotherapy university answer **Assessment** The physiotherapist should assess • Level of consciousness • Motor power • Muscle tone • Sensation • Balance and coordination • Cranial nerve related deficits • Respiratory status • Functional mobility • Activities of daily living **Physiotherapy management** • Appropriate positioning and pressure area care • Respiratory assessment and treatment when clinically indicated • Passive/active ROM exercises • Prevention of contractures and muscle weakness • Early and progressive mobilization when medically stable • Bed mobility and transfer training • Sitting and standing balance exercises • Gait training • Progressive strengthening and endurance training • Functional and ADL rehabilitation • Patient and caregiver education Treatment must be modified according to neurological status surgical precautions intracranial pressure concerns and medical stability //////////////////////////////////// **Decompressive Craniectomy in Traumatic Brain Injury TBI** - -- **Definition** • Decompressive craniectomy DC is a neurosurgical procedure in which a large portion of the skull is removed and the bone flap is not immediately replaced creating additional intracranial space to reduce raised intracranial pressure ICP and prevent or treat brain herniation • It may be performed as part of the surgical treatment of severe traumatic brain injury particularly when intracranial hypertension remains refractory to appropriate medical treatment --- **1 Objectives** • Reduce intracranial pressure ICP • Increase intracranial compliance • Improve cerebral perfusion • Reduce secondary brain injury • Prevent cerebral herniation • Prevent further neurological deterioration • Reduce mortality in appropriately selected patients --- **2 Indications in TBI** • Severe TBI with refractory intracranial hypertension despite appropriate medical treatment • Severe cerebral edema with raised ICP • Significant traumatic brain swelling • Acute subdural hematoma with severe brain swelling or brain bulging after evacuation • Intracranial hematoma associated with significant mass effect • Neurological deterioration due to raised ICP • Clinical or radiological evidence of impending brain herniation • Diffuse cerebral edema with medically refractory intracranial hypertension **Important examination point** • Secondary decompressive craniectomy for late refractory ICP elevation is recommended to improve mortality and favorable outcomes • Current guidelines do not recommend routine early secondary DC simply for modest ICP elevation that is still medically manageable --- **3 Types** **A Primary decompressive craniectomy** • Performed during the initial surgery for a traumatic mass lesion • The bone flap is left out after evacuation of the hematoma or other traumatic lesion • Commonly considered with severe acute subdural hematoma and significant cerebral swelling **B Secondary decompressive craniectomy** • Performed later when intracranial pressure remains refractory to medical treatment • May be unilateral or bifrontal depending on the pattern of brain swelling --- **4 Surgical Steps** 1 General anesthesia • Patient is anesthetized and appropriately positioned 2 Scalp incision • A large scalp incision is made according to the planned decompression 3 Creation of burr holes • Multiple burr holes are created in the skull 4 Removal of large bone flap • A large portion of the skull is removed • In severe unilateral injury a large frontotemporoparietal decompression may be performed 5 Evacuation of traumatic lesion • Hematoma such as subdural or epidural hematoma may be evacuated when indicated 6 Opening of dura • The dura mater is opened widely to allow the swollen brain to expand 7 Duraplasty • The dura may be enlarged using a graft to provide

additional space 8 Hemostasis • Bleeding is controlled and the brain is carefully inspected 9 Bone flap not replaced • Unlike craniotomy the removed bone flap is not immediately replaced 10 Closure • The scalp is closed over the decompressed area 11 Later cranioplasty • After the acute phase has resolved the skull defect is usually reconstructed by cranioplasty when appropriate --- **5 Complications Early complications** • Intracranial bleeding or hematoma expansion • Cerebral edema • External cerebral herniation • CSF leak • Subdural hygroma • Wound infection • Seizures • Hydrocephalus • Contralateral hematoma **Late complications** • Syndrome of the trephined/sinking skin flap syndrome • Hydrocephalus • Persistent subdural collections • Neurological deterioration • Cosmetic skull defect • Complications related to subsequent cranioplasty --- **6 Physiotherapy Implications Assessment** • Level of consciousness/GCS • Pupillary response as documented by the medical team • Motor power • Muscle tone • Sensory function • Balance and coordination • Respiratory status • Vital signs and SpO₂ • Functional mobility **Physiotherapy management** • Proper positioning and pressure area care • Respiratory assessment and treatment when indicated • Passive and active ROM exercises • Prevention of contractures • Prevention of pressure injuries • Gradual mobilization when medically stable • Bed mobility and transfer training • Sitting and standing balance exercises • Gait training • Progressive strengthening and functional rehabilitation • Family/caregiver education **Precautions** • Avoid pressure or trauma to the craniectomy defect • Appropriate head protection should be used according to neurosurgical recommendations • Avoid treatment that may significantly increase ICP • Monitor neurological status and vital signs closely • Stop treatment and seek medical review if there is acute neurological deterioration vomiting reduced consciousness seizures or significant cardiorespiratory instability
//////////////////// **Laminectomy Definition Indications Procedure and Complications** --- **Definition** • Laminectomy is a spinal surgical procedure in which part or all of the lamina of a vertebra is removed to decompress the spinal cord or nerve roots • It is commonly performed to relieve pressure caused by spinal stenosis disc disease tumors or other compressive lesions --- **1 Indications** Laminectomy may be indicated in • Spinal canal stenosis causing neurological symptoms • Compression of spinal cord or nerve roots • Herniated intervertebral disc in selected cases • Severe radiculopathy or sciatica due to compression • Spinal tumors • Epidural abscess or other space occupying lesions • Traumatic spinal cord compression • Some cases of degenerative spinal disease • Cauda equina compression when urgent decompression is required • Persistent neurological symptoms not responding adequately to conservative treatment --- **2 Procedure / Surgical Steps A Pre operative preparation** • Patient assessment and informed consent • Appropriate imaging such as X ray CT or MRI • General anesthesia is usually administered • Patient is positioned prone on the operating table **B Surgical procedure** 1 Skin incision • A midline incision is made over the affected vertebral levels 2 Exposure • Paraspinal muscles are carefully separated from the vertebral structures • The lamina and spinous processes are exposed 3 Identification of affected level • The surgeon confirms the appropriate spinal level 4 Removal of lamina • Part or all of the lamina is removed using appropriate surgical instruments • This creates additional space within the spinal canal 5 Decompression • Compressed spinal cord or nerve roots are decompressed • Additional procedures such as discectomy or foraminotomy may be performed when indicated 6 Hemostasis • Bleeding is controlled 7 Closure • Muscles and soft tissues are returned to position • The wound is closed in layers 8 Post operative care • Neurological status pain wound condition and mobility are monitored • Early mobilization is encouraged when medically appropriate **Flowchart** Anesthesia Prone position Skin incision Muscle exposure Identify vertebral level Remove lamina Decompress neural structures Hemostasis Wound closure Rehabilitation --- **3 Complications A Neurological complications** • Nerve root injury • Spinal cord injury • New weakness or paralysis • Sensory disturbances • Persistent or recurrent radicular pain **B Surgical complications** • Bleeding/hematoma • Infection • Dural tear and CSF leak • Wound problems • Injury to surrounding tissues **C Musculoskeletal complications** • Spinal instability • Post operative back pain • Muscle weakness • Reduced spinal mobility • Recurrent spinal stenosis • Adjacent segment degeneration in some patients **D Other complications** • Deep vein thrombosis • Pulmonary complications related to prolonged immobility • Need for further spinal surgery --- **4 Physiotherapy Implications** Although not specifically requested this earns useful marks in a physiotherapy surgery exam **Assessment** • Pain and neurological symptoms • Muscle strength • Sensory status • ROM • Posture • Balance • Functional mobility • Gait • Respiratory and cardiovascular status **Physiotherapy management** • Proper positioning • Breathing and circulation exercises • Early mobilization when medically stable • Bed mobility and transfer training • Gentle ROM exercises • Progressive strengthening • Gait training • Postural education • Gradual functional rehabilitation • Patient education regarding spinal precautions and safe body mechanics **Precautions** • Avoid excessive spinal flexion extension rotation or loading during the early healing period according to surgical instructions • Avoid heavy lifting until medically cleared • Protect the surgical wound • Monitor for new weakness sensory changes increasing pain fever or wound discharge and report appropriately
//////////////////// **Aneurysm Clipping vs Coiling** --- **Definition of Cerebral Aneurysm** • A cerebral aneurysm is a localized abnormal dilatation or ballooning of a blood vessel in the brain due to weakness of the arterial wall • Rupture can cause subarachnoid hemorrhage SAH which is a neurological emergency --- **1 Aneurysm Clipping Definition** • Surgical clipping is an open neurosurgical procedure in which a small metal clip is placed across the neck of the aneurysm to prevent blood

from entering the aneurysm **Surgical Steps** 1 General anesthesia is given 2 A scalp incision is made 3 A craniotomy is performed 4 The surgeon exposes the aneurysm 5 The parent artery and aneurysm neck are identified 6 A small metal clip is placed across the aneurysm neck 7 Blood flow into the aneurysm is stopped 8 Hemostasis is achieved 9 The bone flap is replaced and the wound is closed **Advantages** • Provides direct visualization of the aneurysm • Usually provides durable occlusion • Lower risk of aneurysm recurrence compared with coiling in many cases • Useful for some complex aneurysms and aneurysms with anatomy unfavorable for endovascular treatment **Disadvantages / Complications** • Requires craniotomy • More invasive • Longer recovery • Risk of brain or blood vessel injury • Intracranial bleeding • Stroke or neurological deficit • Seizures • Infection • CSF leak --- **2 Aneurysm Coiling Definition** • Endovascular coiling is a minimally invasive procedure in which detachable coils are introduced through a catheter into the aneurysm causing thrombosis and preventing blood flow into the aneurysm **Surgical Steps** 1 Local or general anesthesia is administered 2 A catheter is introduced through an artery commonly through the groin or wrist 3 The catheter is guided through the arterial system to the cerebral circulation 4 The aneurysm is identified using angiography 5 A microcatheter is passed into the aneurysm 6 Soft platinum coils are deployed inside the aneurysm 7 Coils promote clot formation within the aneurysm 8 Blood flow into the aneurysm is reduced or stopped 9 The catheter is removed 10 The access site is closed and monitored **Advantages** • Minimally invasive • No craniotomy is required • Generally shorter recovery than surgical clipping • Particularly useful when open surgery carries greater risk • Endovascular treatment can be advantageous in selected patients with ruptured aneurysms **Disadvantages / Complications** • Aneurysm may recanalize or recur • May require repeat treatment and follow up imaging • Thromboembolic complications • Vessel injury • Intracranial hemorrhage • Stroke or neurological deficit • Contrast related complications • Some wide neck aneurysms may require balloon or stent assisted techniques --- **4 Factors Affecting Choice of Treatment** The decision between clipping and coiling depends on • Aneurysm size • Aneurysm location • Shape and neck width • Ruptured or unruptured status • Patients age • General medical condition • Presence of brain swelling or vasospasm • Accessibility of the aneurysm • Experience of the treating neurosurgical/endovascular team --- **5 Physiotherapy Implications** • Assess • Level of consciousness • Motor power • Sensory function • Muscle tone • Balance and coordination • Respiratory status • Functional mobility • Maintain appropriate positioning • Provide respiratory physiotherapy when clinically indicated • Perform ROM exercises to prevent stiffness and contractures • Begin early and progressive mobilization when medically stable • Provide balance gait and functional training • Monitor neurological status during treatment • Educate patient and family regarding safe mobility and rehabilitation /////////////// **Postoperative Complications Shock DVT Wound Infection Atelectasis Pneumonia UTI and Pressure Sores** --- **Definition** Postoperative complications are unwanted conditions that develop after a surgical procedure and may delay recovery impair normal body function increase hospital stay or threaten life --- **1 Shock Definition** • Shock is a state of inadequate tissue perfusion and oxygen delivery resulting in cellular dysfunction and organ failure **Causes after surgery** • Hemorrhage • Severe fluid loss • Sepsis • Cardiac dysfunction • Anaphylaxis **Clinical features** • Hypotension • Tachycardia • Cold clammy skin • Rapid breathing • Reduced urine output • Confusion or reduced consciousness **Physiotherapy implications** • Do not mobilize an unstable patient • Monitor BP HR respiratory rate and SpO2 • Position appropriately according to the medical condition • Once stabilized gradually initiate mobility and rehabilitation --- **2 Deep Vein Thrombosis DVT Definition** • DVT is the formation of a blood clot within a deep vein most commonly in the lower limb **Risk factors** • Prolonged immobilization • Major surgery • Reduced mobility • Dehydration • Previous DVT • Cancer **Clinical features** • Unilateral leg swelling • Calf pain or tenderness • Warmth • Redness • Increased calf circumference **Physiotherapy implications** • Encourage early mobilization when medically appropriate • Ankle pump and active limb exercises • Avoid vigorous massage of a suspected DVT • If DVT is suspected stop treatment and seek medical assessment • Anticoagulation and mechanical prophylaxis may be prescribed according to risk --- **3 Wound Infection Definition** • Wound infection is infection of a surgical incision or wound by microorganisms causing local and/or systemic inflammatory response **Clinical features** • Increased pain • Redness • Swelling • Warmth • Purulent discharge • Fever • Delayed wound healing **Physiotherapy implications** • Inspect the wound before treatment • Maintain strict hand hygiene and infection control precautions • Avoid direct pressure or vigorous massage over an infected wound • Modify or postpone treatment if the patient is systemically unwell • Maintain mobility of unaffected areas • Follow wound care and surgical precautions --- **4 Atelectasis Definition** • Atelectasis is partial or complete collapse of alveoli resulting in reduced ventilation of the affected lung area **Causes** • Shallow breathing due to pain • Anesthesia • Prolonged bed rest • Retained secretions • Abdominal or thoracic surgery **Clinical features** • Breathlessness • Reduced oxygen saturation • Reduced chest expansion • Decreased breath sounds • Tachypnea **Physiotherapy management** • Upright positioning • Deep breathing/thoracic expansion exercises • Supported coughing • Early mobilization • Airway clearance techniques when clinically indicated • Monitor SpO2 and respiratory status --- **5 Pneumonia Definition** • Pneumonia is an infection causing inflammation of the lung parenchyma/alveoli often with consolidation and impaired gas exchange **Clinical features** • Fever • Cough • Sputum production • Breathlessness • Chest discomfort • Reduced oxygen saturation • Fatigue **Physiotherapy implications** • Positioning to optimize ventilation •

Breathing exercises • Airway clearance techniques when indicated • Early mobilization • Progressive walking and exercise • Monitor respiratory rate HR and SpO2 --- **6 Urinary Tract Infection UTI Definition** • UTI is an infection affecting any part of the urinary tract including the bladder urethra ureters or kidneys **Postoperative risk factors** • Urinary catheterization • Prolonged catheter use • Immobility • Inadequate hydration • Urinary retention **Clinical features** • Burning or pain during urination • Increased urinary frequency • Urgency • Lower abdominal discomfort • Fever in more significant infection • Cloudy or foul smelling urine **Physiotherapy implications** • Encourage early mobilization when appropriate • Encourage adequate mobility and hydration according to medical advice • Maintain catheter safety during transfers and walking • Maintain infection control measures • Monitor for fever or systemic deterioration --- **7 Pressure Sores / Pressure Injuries Definition** • A pressure injury is localized damage to the skin and/or underlying tissue usually occurring over a bony prominence due to prolonged pressure or pressure combined with shear **Risk factors** • Prolonged bed rest • Reduced consciousness • Poor nutrition • Incontinence • Reduced sensation • Immobility **Common sites** • Sacrum • Heels • Greater trochanters • Elbows • Occiput **Physiotherapy management** • Regular repositioning • Pressure relieving positions and surfaces • Early mobilization • ROM exercises • Reduce prolonged pressure over bony prominences • Strengthening and functional mobility • Patient and caregiver education • Coordinate with the wound care team

////////// **Physiotherapy Management After Surgery --- Definition** • Physiotherapy management after surgery is the planned use of physiotherapy assessment exercises positioning mobilization and education to prevent postoperative complications restore physical function and promote safe recovery --- **Main Physiotherapy Management 1 Assessment** • Check vital signs HR BP RR and SpO2 • Assess pain • Assess respiratory status • Assess ROM and muscle strength • Assess bed mobility transfers and gait • Check surgical wound and precautions **2 Respiratory Physiotherapy** • Deep breathing exercises • Thoracic expansion exercises • Supported coughing • Airway clearance techniques when indicated • Appropriate positioning • Helps prevent atelectasis and pneumonia **3 Circulatory Exercises** • Ankle pump exercises • Active limb movements • Early mobilization • Helps reduce venous stasis and risk of DVT **4 Early Mobilization** • Bed mobility • Sitting at bedside • Sit to stand • Transfer training • Progressive walking • Prevents deconditioning weakness and pressure injuries **5 ROM and Strengthening** • Passive/active ROM as appropriate • Isometric exercises when indicated • Progressive strengthening • Prevents stiffness muscle weakness and contractures **6 Patient Education** • Teach safe bed mobility and log rolling when appropriate • Wound protection during coughing and movement • Posture and body mechanics • Activity progression and home exercises • Educate about warning signs requiring medical review **Goals** • Prevent postoperative complications • Reduce pain and improve mobility • Maintain muscle strength and ROM • Improve respiratory and cardiovascular function • Restore functional independence and ADLs ////////// **Mastectomy Complications and Physiotherapy Management --- 1 Definition of Mastectomy** • Mastectomy is a surgical procedure in which part or all of the breast tissue is removed most commonly for the treatment of breast cancer • It may be performed on one breast unilateral or both breasts bilateral **Types** • Simple/total mastectomy removal of the whole breast • Modified radical mastectomy removal of the breast with axillary lymph node dissection • Skin sparing mastectomy breast tissue is removed while preserving most of the breast skin • Nipple sparing mastectomy breast tissue is removed while preserving the nipple areola complex • Radical mastectomy extensive removal of breast chest wall muscles and lymph nodes now rarely performed --- **2 Complications After Mastectomy Early complications** • Pain at the surgical site • Wound infection • Bleeding or hematoma • Seroma which is accumulation of fluid around the surgical site • Wound healing problems • Reduced shoulder movement • Arm and shoulder stiffness • Numbness or altered sensation around the chest or arm • Fatigue **Late complications** • Lymphedema of the arm hand breast or chest wall particularly after lymph node removal or radiotherapy • Shoulder dysfunction or frozen shoulder • Post mastectomy pain syndrome • Scar adhesion and tissue tightness • Axillary web syndrome cording • Postural abnormalities especially rounded shoulders • Muscle weakness and reduced physical activity • Reduced functional independence • Psychological problems related to altered body image --- **3 Physiotherapy Management After Mastectomy A Physiotherapy Assessment** • Assess pain and surgical wound • Assess shoulder and upper limb ROM • Assess muscle strength • Assess posture • Assess sensation • Assess upper limb swelling and signs of lymphedema • Assess respiratory status • Assess functional activities and ADLs --- **B Respiratory Physiotherapy** • Deep breathing exercises • Thoracic expansion exercises • Supported coughing when required • Appropriate positioning • Early mobilization • Helps prevent respiratory complications after surgery --- **C Shoulder and Upper Limb Exercises** • Begin gentle exercises according to surgical instructions • Shoulder flexion exercises • Shoulder abduction exercises • Elbow flexion and extension • Wrist and finger movements • Scapular movements • Gradually restore full shoulder ROM Breast surgery rehabilitation commonly includes postoperative shoulder exercises to maintain or restore arm mobility --- **D Lymphedema Prevention and Management** • Encourage regular gradual arm and shoulder movement • Avoid prolonged inactivity of the affected arm • Maintain healthy body weight and regular physical activity • Protect the skin from cuts burns and infection • Monitor for • Arm swelling • Heaviness • Tightness • Reduced ROM • Refer to a specialist lymphedema service if lymphedema develops For established lymphedema decongestive lymphatic therapy may include exercise compression skin

care and specialist manual lymphatic drainage --- **E Postural Correction** • Encourage upright sitting and standing • Avoid rounded shoulders • Perform scapular retraction exercises • Maintain normal thoracic mobility • Correct protective postures caused by pain Good posture helps maintain shoulder function and may reduce muscular tension --- **F Scar Management** • Once the wound is fully healed scar mobility can be assessed • Gentle scar mobilization may be used when appropriate • Stretching exercises can help reduce tissue tightness • Monitor for excessive scar formation or adhesions --- **G Functional Rehabilitation** • Bed mobility training • Transfer training • Progressive walking • Upper limb functional activities • Progressive strengthening • Gradual return to ADLs • Gradual return to work exercise and recreational activities according to medical clearance --- **H Patient Education** • Follow wound care instructions • Avoid heavy lifting and strenuous activity until medically cleared • Perform prescribed exercises regularly • Protect the affected arm from cuts burns and infection • Maintain good skin care • Recognize early signs of lymphedema • Maintain regular physical activity • Seek medical advice for increasing swelling redness wound discharge or severe pain --- **4 Physiotherapy Goals** • Reduce pain and stiffness • Restore shoulder and upper limb ROM • Maintain muscle strength • Prevent or manage lymphedema • Prevent postural abnormalities • Improve functional independence • Prevent complications of immobility • Improve quality of life and return to normal activities //////////////// **Amputation Definition Indications Levels Complications Stump Care Phantom Limb Prosthetic Rehabilitation and Cineplasty** --- **1 Definition** • Amputation is the surgical removal of part or all of a limb or extremity because of disease trauma or other conditions in which the limb cannot be preserved • The remaining portion of the limb is called the residual limb or stump --- **2 Indications of Amputation** • Severe trauma with an unsalvageable limb • Severe peripheral vascular disease/ischemia • Diabetic foot with irreversible tissue damage • Gangrene • Severe infection or uncontrolled osteomyelitis • Malignant bone or soft tissue tumour • Congenital limb deformity in selected cases • Severe burns or frostbite with non viable tissue • Failed previous reconstruction or repeated infection --- **3 Levels of Amputation Lower limb** • Hemipelvectomy • Hip disarticulation • Transfemoral amputation above knee • Knee disarticulation • Transtibial amputation below knee • Symes amputation • Ankle/foot amputation • Partial foot/toe amputation **Upper limb** • Forequarter amputation • Shoulder disarticulation • Transhumeral amputation above elbow • Elbow disarticulation • Transradial amputation below elbow • Wrist disarticulation • Partial hand/finger amputation **Principle** Preserve the maximum useful limb length and whenever possible preserve important joints to optimize functional and prosthetic outcomes --- **4 Complications of Amputation Early complications** • Hemorrhage • Hematoma • Infection • Wound breakdown • Edema • Skin necrosis • Pain • Poor wound healing **Late complications** • Joint contracture • Muscle weakness • Residual limb pain • Neuroma • Phantom limb sensation • Phantom limb pain • Poor prosthetic fit • Skin problems due to prosthetic pressure • Reduced mobility and functional independence --- **5 Stump / Residual Limb Care** • Inspect the stump daily for redness swelling pressure areas and skin breakdown • Maintain good stump hygiene • Keep the healed residual limb clean and dry • Control edema using appropriate compression/shrinker techniques when prescribed • Shape the stump appropriately for prosthetic fitting • Use correct positioning to prevent contractures • Perform regular ROM exercises • Strengthen the residual limb and remaining muscles • Perform desensitization when appropriate • Care for the scar after adequate wound healing • Avoid excessive pressure or friction • Protect the opposite limb particularly in patients with vascular disease or diabetes --- **6 Phantom Limb Definition** • Phantom limb sensation is the feeling that the amputated limb is still present • Phantom limb pain PLP is painful sensation perceived in the missing limb **Features** • Burning pain • Shooting or electric like pain • Cramping • Tingling • Feeling of movement or position of the missing limb **Management** • Patient education and reassurance • Desensitization techniques • Gentle tapping and massage of the healed residual limb • Graded motor imagery • Mirror therapy may be used as part of rehabilitation • Appropriate prosthetic use when suitable • Psychological support when required • Medical management of pain when necessary --- **7 Prosthetic Rehabilitation A Pre prosthetic phase** • Control edema • Shape the residual limb • Maintain ROM • Prevent contractures • Strengthen residual and intact limbs • Improve balance • Improve cardiovascular endurance • Desensitize the residual limb • Train transfers and mobility • Teach ADLs **B Prosthetic fitting** • Assess residual limb condition • Select an appropriate prosthesis • Ensure proper socket fit • Teach donning and doffing • Teach prosthetic hygiene and skin inspection • Gradually increase wearing time **C Prosthetic training** • Weight shifting • Standing balance • Parallel bar training • Gait training • Walking with assistive devices initially when required • Stair training • Uneven surface training • Functional and community mobility **D Follow up** • Regular assessment of skin and residual limb • Prosthetic adjustment as limb volume changes • Gait assessment • Maintenance of prosthetic components • Long term functional and psychological support --- **8 Cineplasty Definition** • Cineplasty is a surgical technique in which a muscle tendon unit of the residual limb is brought through the stump and attached to a mechanical control system allowing the patients muscle contraction to operate or control a prosthetic device **Principle** Muscle contraction movement of control mechanism prosthetic movement **Advantages** • Provides more direct control of the prosthesis • Provides sensory feedback through muscle activity • May improve functional control of some prosthetic devices • Can be particularly useful in selected upper limb amputees --- **9 Physiotherapy Goals After Amputation** • Reduce pain and edema • Prevent contractures • Maintain

and improve muscle strength • Improve balance and coordination • Develop independence in transfers and ADLs • Prepare the residual limb for prosthetic fitting • Train safe prosthetic use • Restore gait and functional mobility • Promote psychological adjustment and community reintegration

//////////////////// **Vascular Disorders Varicose Veins Peripheral Arterial Occlusive Disease DVT and Surgical Management** --- **1 Varicose Veins Definition** • Varicose veins are dilated elongated and tortuous superficial veins most commonly in the lower limbs caused by failure of venous valves leading to venous reflux and pooling of blood

Clinical features • Visible twisted and enlarged veins • Leg pain or aching • Heaviness and fatigue of legs • Ankle swelling • Itching • Skin pigmentation or eczema in chronic disease • Venous ulceration in severe cases • Superficial thrombophlebitis may occur **Management** • Regular exercise and weight management • Leg elevation • Compression therapy when appropriate • Endothermal ablation • Ultrasound guided foam sclerotherapy • Surgical ligation/stripping when other interventions are unsuitable --- **2 Peripheral Arterial Occlusive Disease PAOD Definition** •

Peripheral arterial occlusive disease is a condition in which arterial blood flow to the limbs is reduced due to narrowing or blockage of peripheral arteries most commonly because of atherosclerosis **Clinical features** • Intermittent claudication pain in the limb during walking that improves with rest • Cold extremity • Pale or bluish skin • Reduced or absent peripheral pulses • Shiny thin skin • Loss of hair • Delayed wound healing • Ischemic ulcers • Severe disease may cause rest pain and gangrene **Management** • Smoking cessation • Regular/supervised exercise • Control of diabetes hypertension and cholesterol • Antiplatelet and lipid lowering medication as prescribed • Angioplasty/stenting for selected lesions • Arterial bypass surgery when indicated • Amputation may be required for irreversible tissue loss/gangrene --- **3 Deep Vein Thrombosis DVT Definition** • DVT is the formation of a blood clot within a deep vein usually in the lower limb thigh or

pelvis **Risk factors** • Prolonged immobilization • Major surgery • Trauma • Cancer • Previous DVT • Pregnancy • Obesity • Prolonged sitting or bed rest **Clinical features** • Unilateral leg swelling • Calf or thigh pain/tenderness • Warmth • Redness or discoloration • Increased limb circumference • Some patients may have no symptoms **Complications** • Pulmonary embolism PE • Post thrombotic syndrome • Chronic swelling and pain **Management** • Anticoagulant therapy is the main treatment • Compression may help selected patients with symptoms • Leg elevation for swelling • Early walking/mobilization when medically appropriate • Selected extensive DVTs may require catheter based clot removal **Physiotherapy precaution** • Do not perform vigorous massage over a suspected DVT • Suspected DVT requires prompt medical assessment • Mobilization should follow the treating teams instructions --- **4 Surgical Management of Vascular Disorders** Surgical treatment depends on whether the disorder is venous or arterial **A Varicose veins** • Endovenous thermal ablation • Ultrasound guided foam sclerotherapy • Surgical ligation and stripping when indicated • Phlebectomy for selected tributaries **B Peripheral arterial occlusive disease** • Angioplasty • Balloon catheter is used to widen the narrowed artery • Stent placement • A stent may be inserted to maintain arterial patency • Arterial bypass grafting • A graft creates an alternative route around the blocked artery • Thrombectomy/embolectomy may be performed for selected acute arterial occlusions • Amputation may be required when severe ischemia has resulted in irreversible tissue loss **C Aneurysms** • Endovascular stent graft repair • Open surgical aneurysm repair **D Severe DVT** • Anticoagulation is generally first line treatment • Selected patients with extensive iliofemoral DVT may undergo catheter directed thrombolysis or thrombectomy --- **5 Physiotherapy Implications** • Assess peripheral pulses skin condition pain swelling and functional mobility • Encourage graded exercise in stable peripheral arterial disease • Promote early mobilization after vascular surgery when medically appropriate • Provide ankle and lower limb exercises • Monitor wound healing • Educate regarding foot and skin care • Avoid excessive pressure or trauma to ischemic tissues • Avoid vigorous massage when DVT is suspected • After amputation provide stump care strengthening balance and prosthetic rehabilitation

//////////////////// **Wound Wound Healing Factors Types Non Healing Wounds and Surgical Wound Management** --- **1 Definition of Wound** • A wound is a disruption or break in the normal continuity and integrity of the skin mucous membrane or underlying tissues caused by trauma surgery disease or other factors • A wound may be open or closed acute or chronic --- **2 Wound Healing Definition** • Wound healing is the natural biological process by which the body repairs damaged tissue and restores tissue continuity and function **Phases of Wound Healing** 1 Hemostasis • Occurs immediately after injury • Vasoconstriction occurs • Platelets aggregate • Fibrin clot is formed • Stops bleeding 2 Inflammation • Neutrophils and macrophages enter the wound • Removes bacteria dead tissue and debris • Produces inflammatory mediators 3 Proliferation • Fibroblasts produce collagen • Granulation tissue develops • New blood vessels form by angiogenesis • Re epithelialization occurs • Wound contraction occurs 4 Remodeling/Maturation • Collagen is reorganized • Scar tissue matures • Tensile strength gradually increases --- **3 Factors Affecting Wound Healing A Local Factors** • Infection • Poor blood supply/ischemia • Hypoxia • Excessive edema • Wound size and depth • Foreign bodies • Repeated trauma or pressure • Excessive wound tension • Poor wound hygiene • Tissue necrosis **B Systemic Factors** • Age • Malnutrition and protein deficiency • Vitamin and mineral deficiencies • Diabetes mellitus • Anemia • Smoking • Obesity • Immunosuppression • Poor circulation • Systemic infection • Corticosteroid therapy • Chemotherapy and radiotherapy --- **4 Types of Wounds A According to duration** • Acute wound • Develops suddenly • Usually progresses through normal

healing • Example surgical incision laceration • Chronic/non healing wound • Fails to progress normally through healing • Example diabetic ulcer venous ulcer pressure ulcer **B According to cause** • Surgical wound • Traumatic wound • Burn wound • Pressure wound • Diabetic wound • Vascular ulcer **C According to skin integrity** • Open wound • Skin is broken • Example laceration abrasion • Closed wound • Skin remains intact but underlying tissues are damaged • Example contusion or hematoma **D According to contamination** • Clean wound • Clean contaminated wound • Contaminated wound • Dirty/infected wound --- **5 Non Healing Wounds Definition** • A non healing or chronic wound is a wound that fails to progress through the normal healing process and remains open or recurrent • Chronic wounds commonly include diabetic ulcers venous ulcers arterial ulcers and pressure ulcers **Causes** • Diabetes mellitus • Peripheral arterial disease • Venous insufficiency • Pressure • Infection • Poor nutrition • Smoking • Poor oxygenation • Edema • Neuropathy • Immunosuppression • Repeated trauma **Clinical features** • Persistent wound • Delayed or absent epithelialization • Excessive or abnormal discharge • Slough or necrotic tissue • Poor granulation tissue • Recurrent infection • Wound enlargement --- **6 Surgical Wound Management A Assessment** • Assess wound location and size • Observe wound edges • Check colour and condition of wound bed • Assess exudate/discharge • Check for redness swelling warmth and pain • Assess for wound dehiscence • Monitor temperature and systemic signs of infection **B Wound Care** • Maintain aseptic technique during dressing changes • Keep the wound appropriately protected • Use appropriate dressings according to wound type • Clean the wound according to clinical protocol • Sterile saline is recommended for cleansing surgical wounds in the early postoperative period according to NICE guidance **C Infection Prevention** • Proper hand hygiene • Aseptic non touch technique • Appropriate wound dressing • Early recognition of infection • Antibiotics when clinically indicated for surgical site infection • Report increasing redness pain swelling purulent discharge or fever **D Promote Healing** • Maintain adequate nutrition and hydration • Maintain adequate oxygenation and circulation • Control blood glucose in diabetic patients • Reduce pressure on the wound • Control edema • Encourage appropriate physical activity and mobility **E Physiotherapy Management** • Appropriate positioning • Pressure relieving techniques • Early mobilization when medically safe • ROM exercises • Muscle strengthening • Breathing exercises after major surgery • Circulatory exercises • Functional mobility and gait training • Patient education regarding wound protection • Scar management only after adequate wound closure/healing --- **7 Complications of Wounds** • Wound infection • Hematoma • Seroma • Wound dehiscence • Necrosis • Delayed healing • Chronic ulceration • Excessive scar formation • Contracture • Sepsis in severe infection --- **Flowchart of Wound Healing** Wound/Injury Hemostasis Inflammation Proliferation Remodeling/Maturation Scar formation and tissue strengthening // **Fluid and Electrolytes** --- **1 Definition of Fluid and Electrolytes** • Body fluid is the water present in the body that forms the major component of blood cells and extracellular spaces • Electrolytes are minerals that carry an electrical charge when dissolved in body fluids • Important electrolytes include • Sodium Na • Potassium K • Calcium Ca • Chloride Cl • Magnesium Mg • Phosphate PO4 --- **2 Fluid Balance** • Fluid balance is the maintenance of a relatively stable relationship between fluid intake and fluid output • Fluid intake drinking food enteral feeding and IV fluids • Fluid output urine sweat stool respiration and other losses • Normal fluid balance is important for • Blood circulation • Blood pressure • Cellular function • Temperature regulation • Electrolyte balance --- **3 Dehydration** • Dehydration is a condition in which the body loses more water than it takes in resulting in insufficient body water • Causes • Vomiting • Diarrhea • Excessive sweating • Fever • Inadequate fluid intake • Blood/fluid loss • Clinical features • Thirst • Dry mouth • Reduced urine output • Dark urine • Dizziness • Weakness • Tachycardia • Hypotension in severe cases --- **4 Electrolyte Imbalance** • Electrolyte imbalance is an abnormal increase or decrease in the concentration of one or more electrolytes in body fluids • Common abnormalities • Hyponatremia / hypernatremia • Hypokalemia / hyperkalemia • Hypocalcemia / hypercalcemia • Hypomagnesemia / hypermagnesemia • Clinical features • Muscle weakness • Cramps • Fatigue • Confusion • Nausea/vomiting • Abnormal heart rhythm particularly with significant potassium abnormalities --- **5 Management** • Identify and treat the underlying cause • Monitor • Fluid intake/output • Blood pressure • Heart rate • Urine output • Serum electrolytes • Mild dehydration oral fluids or oral rehydration solution when appropriate • Severe dehydration IV fluid replacement usually with appropriate isotonic crystalloid depending on the clinical situation • Replace deficient electrolytes appropriately • Restrict fluids when clinically indicated for specific fluid overload conditions • Avoid rapid or excessive correction of electrolyte abnormalities • Monitor the patients clinical response **Physiotherapy Precaution** • Check the patients vital signs and medical stability before treatment • Modify or postpone intensive physiotherapy in significant dehydration or electrolyte disturbance • Encourage appropriate mobility and hydration according to the medical plan // **Drains and Tubes ICD Foley Catheter Hemovac CVC and PICC** -- **1 Drains Definition** • A surgical drain is a tube or device inserted into a body cavity wound or surgical site to remove blood pus serous fluid bile or other unwanted collections • Drains help prevent fluid accumulation and may reduce complications such as infection or hematoma --- **2 ICD Intercostal Drain Definition** • An intercostal drain ICD is a tube inserted through the chest wall into the pleural cavity to remove air blood or fluid and allow the lung to re expand **Indications** • Pneumothorax • Hemothorax • Pleural effusion • Empyema • Post thoracic surgery **Physiotherapy precautions** • Check the

drain system before treatment • Keep the drainage bottle/system upright and below chest level • Avoid pulling kinking or compressing the tube • Encourage breathing exercises and early mobilization when medically stable • Monitor SpO2 respiratory rate and symptoms --- **3 Foley Catheter Definition** • A Foley catheter is an indwelling urinary catheter inserted through the urethra into the urinary bladder to continuously drain urine • A balloon near the tip keeps the catheter in place **Indications** • Urinary retention • Accurate urine output monitoring in selected critically ill patients • Perioperative bladder drainage • Certain urological conditions **Physiotherapy precautions** • Keep the drainage bag below bladder level • Avoid kinking or pulling the catheter • Secure the catheter during mobilization • Encourage early mobility when appropriate • Maintain infection control precautions --- **4 Hemovac Definition** • Hemovac is a closed suction drainage system used to remove blood and other fluid from a surgical wound **Uses** • Orthopaedic surgery • Joint replacement • Spinal surgery • Other operations where postoperative fluid collection is expected **Physiotherapy precautions** • Check the drain and suction reservoir before treatment • Keep the reservoir secured • Avoid pulling or compressing the tubing • Monitor the amount and appearance of drainage • Mobilize carefully without tension on the drain --- **5 CVC Central Venous Catheter Definition** • A central venous catheter CVC is a catheter whose tip lies in a large central vein commonly the superior vena cava for administration of medications fluids or other therapies **Indications** • Administration of irritating/vesicant medications • Long term IV therapy • Administration of parenteral nutrition • Hemodynamic monitoring in selected patients • Poor peripheral venous access **Physiotherapy precautions** • Ensure the catheter is secure before movement • Avoid excessive traction on the catheter • Monitor the insertion site for redness swelling or bleeding • Observe for signs of infection or catheter complications • Coordinate with nursing/medical staff during mobilization --- **6 PICC Peripherally Inserted Central Catheter Definition** • A PICC is a long flexible catheter inserted through a peripheral vein usually in the upper arm with its tip advanced into a central vein **Indications** • Long term IV antibiotics • Long term medication administration • Chemotherapy • Parenteral nutrition • Repeated blood sampling in selected patients **Physiotherapy precautions** • Keep the PICC secure • Avoid pulling or excessive pressure on the catheter • Follow activity restrictions provided by the medical team • Monitor the arm for • Swelling • Pain • Redness • Bleeding • Stop treatment and report suspected catheter complications //////////////// **Abdominal and Pelvic Surgeries**

Appendectomy Herniorrhaphy Cholecystectomy Hysterectomy Colostomy and Ileostomy Point wise format for Physiotherapy Surgery Exam --- **1 APPENDECTOMY Definition** • Appendectomy is the surgical removal of the vermiform appendix • It is most commonly performed for acute appendicitis **Indications** • Acute uncomplicated appendicitis • Complicated appendicitis • Gangrenous appendix • Perforated appendix • Appendicular abscess in selected cases • Generalized peritonitis due to perforation • Appendiceal neoplasm in selected cases **Procedure / Surgical Steps Laparoscopic appendectomy** 1 General anesthesia is administered 2 Patient is positioned supine 3 Abdomen is prepared and draped 4 Pneumoperitoneum is created using CO2 5 Laparoscopic ports are inserted 6 Cecum and appendix are identified 7 Appendix is mobilized 8 Mesoappendix and appendicular vessels are divided 9 Base of appendix is secured and divided 10 Appendix is removed usually using a retrieval bag 11 Operative site is inspected for bleeding or contamination 12 CO2 is released 13 Ports are removed and wounds are closed **Flow** Anesthesia Pneumoperitoneum Identify appendix Divide mesoappendix Divide appendix Remove appendix Check Close **Complications** • Wound infection • Intra abdominal abscess • Bleeding/hematoma • Postoperative pain • Ileus • Adhesive intestinal obstruction • Peritonitis/sepsis in complicated cases • Stump appendicitis **Physiotherapy** • Assess vital signs pain and respiratory status • Deep breathing exercises • Thoracic expansion exercises • Supported coughing • Ankle pump exercises • Early mobilization • Bed mobility and transfer training • Progressive walking • Gentle ROM exercises • Teach log rolling for bed mobility when appropriate • Protect the abdominal wound • Avoid heavy lifting/strenuous activity until medically cleared --- **2 HERNIORRHAPHY Definition** • Herniorrhaphy is the surgical repair of a hernia by reducing the herniated contents and repairing the abdominal wall defect • Modern repairs frequently use mesh reinforcement when appropriate **Indications** • Symptomatic hernia • Painful hernia • Enlarging hernia • Irreducible/incarcerated hernia • Strangulated hernia • Intestinal obstruction caused by hernia • Recurrent hernia • Risk of future complications in selected patients **Procedure / Surgical Steps** 1 Patient is assessed and prepared 2 Appropriate anesthesia is administered 3 Incision is made over the hernia 4 Hernia sac is identified 5 Sac is dissected from surrounding structures 6 Herniated bowel/omentum is inspected 7 Viable contents are reduced into the abdominal cavity 8 Non viable bowel may require resection 9 Hernia sac is reduced or excised 10 Fascial defect is repaired 11 Mesh may be placed for reinforcement 12 Hemostasis is achieved 13 Wound is closed in layers **Complications** • Hematoma • Seroma • Wound infection • Chronic postoperative pain • Urinary retention • Nerve injury • Injury to bowel bladder or blood vessels • Mesh infection • Recurrence of hernia **Physiotherapy** • Deep breathing exercises • Supported coughing • Early mobilization • Ankle pump exercises • Gentle ROM exercises • Progressive walking • Teach safe bed mobility • Gradual abdominal strengthening after adequate healing and medical clearance • Avoid excessive straining and heavy lifting during the healing period • Educate regarding posture and body mechanics --- **3 CHOLECYSTECTOMY Definition** • Cholecystectomy is the surgical removal of the gallbladder • It is commonly performed

laparoscopically **Indications** • Symptomatic gallstones • Acute cholecystitis • Chronic cholecystitis • Gallstone related complications • Gallstone pancreatitis in appropriate cases • Selected gallbladder polyps or tumours • Gallbladder trauma in selected cases **Procedure / Surgical Steps Laparoscopic cholecystectomy** 1 General anesthesia is administered 2 Patient is positioned appropriately 3 Abdomen is prepared and draped 4 Pneumoperitoneum is created with CO₂ 5 Laparoscopic ports are inserted 6 Gallbladder is exposed and retracted 7 Hepatocystic/Calots triangle is exposed 8 Cystic duct and cystic artery are identified 9 Critical view of safety is obtained 10 Cystic duct and artery are clipped/ligated and divided 11 Gallbladder is dissected from the liver bed 12 Gallbladder is removed through a port 13 Hemostasis and bile leakage are checked 14 Pneumoperitoneum is released 15 Ports are removed and wounds are closed **Complications** • Bleeding • Wound infection • Bile leak • Bile duct injury • Injury to surrounding organs • Retained common bile duct stone • Postoperative nausea/vomiting • Conversion to open surgery when required **Physiotherapy** • Assess respiratory status • Deep breathing and thoracic expansion exercises • Supported coughing • Appropriate positioning • Ankle pump exercises • Early mobilization • Progressive walking • Gentle ROM • Wound protection • Gradual return to functional activities --- **4**

HYSTERECTOMY Definition • Hysterectomy is the surgical removal of the uterus • Depending on the procedure the cervix fallopian tubes and ovaries may also be removed **Indications** • Abnormal uterine bleeding • Uterine fibroids/leiomyoma • Gynecological malignancy • Premalignant disease • Endometriosis • Pelvic organ prolapse • Chronic pelvic pain in selected cases • Pelvic infection in selected cases **Types** • Total hysterectomy • Subtotal/supracervical hysterectomy • Radical hysterectomy • Hysterectomy with removal of tubes and ovaries when indicated **Procedure / Surgical Steps Abdominal Hysterectomy** 1 Patient is assessed and prepared 2 General anesthesia is administered 3 Patient is positioned appropriately 4 Abdomen is prepared and draped 5 Abdominal incision is made 6 Pelvic organs are inspected 7 Uterus is mobilized 8 Supporting ligaments are identified and divided 9 Uterine vessels are ligated and divided 10 Bladder is separated from the lower uterus/cervix 11 Uterus and cervix are removed 12 Vaginal cuff is closed 13 Hemostasis is checked 14 Abdominal wound is closed in layers **Complications** • Hemorrhage • Wound infection • UTI • Pelvic infection/abscess • DVT/VTE • Bladder injury • Ureteric injury • Bowel injury • Ileus • Nerve injury • Vaginal cuff dehiscence • Urinary dysfunction **Physiotherapy** • Assess pain respiratory status and mobility • Deep breathing exercises • Circulatory exercises • Early mobilization • Bed mobility and transfer training • Gentle ROM exercises • Progressive walking • Postural education • Pelvic floor muscle exercises when appropriate • Gradual abdominal strengthening • Avoid heavy lifting until medically cleared • Educate regarding safe return to ADLs --- **5 COLOSTOMY Definition** • Colostomy is a surgical procedure in which a portion of the colon is brought through the abdominal wall to form a stoma allowing feces to leave the body through the stoma • It may be temporary or permanent **Indications** • Colorectal cancer • Bowel obstruction • Bowel perforation • Diverticular disease with obstruction • Inflammatory bowel disease • Traumatic bowel injury • Protection of a distal bowel anastomosis • Gangrenous sigmoid volvulus • Severe perineal/anorectal disease • Anastomotic leak • Fecal diversion **Types** • End colostomy • Loop colostomy • Double barrel colostomy • Temporary or permanent

colostomy **Procedure / Surgical Steps** 1 Patient is assessed and prepared 2 Stoma site is selected and marked 3 General anesthesia is administered 4 Abdomen is accessed by open or laparoscopic technique 5 Diseased bowel is identified 6 Bowel resection is performed if required 7 Appropriate colon segment is selected 8 Abdominal wall opening is created 9 Colon is brought through the opening without excessive tension 10 Colon is opened 11 Bowel edge is sutured to the skin 12 Stoma is matured 13 Abdominal wound is closed 14 Stoma appliance is applied **Complications** • Stoma ischemia/necrosis • Stoma retraction • Stoma prolapse • Stenosis • Peristomal skin irritation • Parastomal hernia • Bleeding • Hematoma • Bowel obstruction/ileus • Wound infection • DVT • Atelectasis/pneumonia **Physiotherapy** • Respiratory exercises • Deep breathing and supported coughing • Early mobilization • Ankle pump exercises • Bed mobility and transfer training • Progressive walking • Avoid direct pressure over the stoma • Protect the stoma appliance during movement • Gentle ROM exercises • Gradual strengthening • Avoid excessive abdominal pressure in the early healing period • Teach safe posture and body mechanics • Coordinate with the stoma care team --- **6 ILEOSTOMY Definition** • Ileostomy is a procedure in which the ileum is brought through the abdominal wall to form a stoma allowing intestinal contents to leave the body through the stoma • It can be temporary or permanent and may be an end or loop ileostomy **Indications** • Ulcerative colitis • Crohns disease • Colorectal cancer • Total colectomy/proctocolectomy • Severe bowel inflammation • Bowel perforation • Bowel ischemia • Protection of a distal intestinal/colorectal anastomosis • Anastomotic leak • Severe perineal disease **Types** • End ileostomy • Loop ileostomy • Temporary ileostomy • Permanent ileostomy **Procedure / Surgical Steps** 1 Patient is assessed and prepared 2 Appropriate stoma site is selected and marked 3 General anesthesia is administered 4 Abdomen is accessed 5 Diseased bowel is identified and resected when required 6 Suitable ileal segment is selected 7 Opening is created in the abdominal wall 8 Ileum is brought through the opening 9 Ileum is checked for adequate blood supply and absence of tension 10 Ileum is opened 11 Stoma is everted/spouted to direct intestinal contents away from the skin 12 Ileum is sutured to the skin 13 Abdominal wound is closed 14 Stoma appliance is applied **Complications** • Dehydration • Fluid and electrolyte imbalance • High output ileostomy • Stoma ischemia/necrosis

• Stoma retraction • Stoma prolapse • Stenosis • Peristomal skin irritation • Parastomal hernia • Bleeding • Obstruction • Infection/abscess • Renal impairment secondary to significant fluid loss **Physiotherapy** • Assess general condition and hydration status • Respiratory exercises • Circulatory exercises • Early mobilization • Bed mobility and transfer training • Progressive walking • Gentle ROM exercises • Protect the stoma during transfers and exercise • Avoid excessive abdominal pressure during early healing • Progressive strengthening • Monitor fatigue and exercise tolerance • Educate regarding safe activity and stoma protection • Coordinate with the stoma care team ///// /////////////////////////////////// **1 CT vs MRI --- CT**

Computed Tomography • Uses X rays and computer processing to produce cross sectional images • Fast and widely available • Excellent for • Acute intracranial hemorrhage • Head trauma • Skull fractures • Bone abnormalities • Uses ionizing radiation • Provides less soft tissue contrast than MRI --- **MRI Magnetic Resonance Imaging** • Uses a strong magnetic field and radiofrequency waves • Does not use ionizing radiation • Provides excellent soft tissue contrast • Better for • Brain and spinal cord lesions • Tumours • Demyelinating disease • Early ischemic stroke • Ligament muscle and joint pathology • More time consuming and may be contraindicated or require special precautions with certain implants --- **2**

Indications of CT and MRI Indications of CT • Head injury and skull fracture • Suspected intracranial hemorrhage • Acute neurological emergencies • Stroke assessment • Bone pathology • Chest and abdominal emergencies • Detection of masses and structural abnormalities **Indications of MRI** • Brain tumours • Spinal cord and nerve lesions • Multiple sclerosis/demyelination • Early ischemic stroke • Joint and soft tissue injuries • Ligament and tendon pathology • Detailed assessment of brain and spinal lesions --- **3 X Ray Interpretation Basics A Check the patient and film** • Confirm patient identity • Check the correct body part and side • Identify the view/projection AP PA lateral etc • Check image quality and positioning **B Use a systematic approach** For a chest X ray a common approach is ABCDE • A Airway • Tracheal position • Airway patency • B Breathing • Lung fields • Pneumothorax • Consolidation • Pleural effusion • C Circulation • Heart size and cardiac silhouette • Mediastinum • D Disability/diaphragm • Diaphragms • Costophrenic angles • Check for free air where relevant • E Everything else • Bones • Soft tissues • Lines tubes and medical devices **Important X ray findings** • Fracture break/discontinuity in bone cortex • Dislocation abnormal relationship of joint surfaces • Consolidation increased lung opacity often associated with infection • Pneumothorax pleural air with absent peripheral lung markings • Pleural effusion fluid causing blunting of the costophrenic angle /////////////////////////////////// **PAOD Peripheral Arterial Occlusive Disease** --- **1 Definition** • Peripheral Arterial Occlusive Disease PAOD is a condition in which arterial blood flow to the peripheral limbs is reduced due to narrowing or blockage of arteries most commonly due to atherosclerosis nhs uk --- **2**

Causes / Risk Factors • Atherosclerosis • Smoking • Diabetes mellitus • Hypertension • Hyperlipidemia • Increasing age • Obesity • Family history of cardiovascular disease --- **3 Clinical Features** • Intermittent claudication pain/cramping in the limb during walking that improves with rest • Cold extremity • Pale or bluish skin • Reduced/absent peripheral pulses • Shiny thin skin • Loss of limb hair • Delayed wound healing • Ischemic ulcers • Severe disease rest pain and gangrene --- **4 Complications** • Chronic ischemia • Non healing ulcers • Tissue necrosis • Gangrene • Limb loss/amputation • Increased cardiovascular risk including heart attack and stroke nhs uk --- **5 Management** • Smoking cessation • Supervised/structured exercise therapy • Control of diabetes blood pressure and cholesterol • Antiplatelet and lipid lowering medication as prescribed • Angioplasty/stenting for selected patients • Arterial bypass surgery for selected severe disease • Amputation may be required for irreversible tissue loss **Physiotherapy Role** • Graded walking/exercise programme as appropriate • Lower limb strengthening • Functional mobility training • Foot and skin care education • Monitor pain skin condition and peripheral circulation /////////////////////////////////// **Surgical Infections Classification** --- **Definition** • Surgical infection is an infection involving a wound tissue or organ that occurs before during or after a surgical procedure and may interfere with normal healing --- **Classification of Surgical Infections 1 According to Clinical Course** • Acute surgical infections • Develop rapidly • Usually associated with acute inflammation • Examples • Cellulitis • Abscess • Acute wound infection • Necrotizing soft tissue infection • Chronic surgical infections • Persist for a prolonged period • May cause fibrosis sinus formation or tissue destruction • Examples • Tuberculosis • Chronic osteomyelitis • Chronic wound infection --- **2 According to Site** • Superficial infections • Involve skin and subcutaneous tissue • Examples cellulitis superficial wound infection • Deep infections • Involve deep fascia muscle or deeper tissues • Examples deep incisional surgical site infection deep abscess • Organ/space infections • Involve an organ or body space related to surgery • Examples • Intra abdominal abscess • Pelvic abscess • Postoperative pneumonia --- **3 According to Source** • Exogenous infection • Organisms enter from outside the patients body • Example contamination from instruments environment or healthcare personnel • Endogenous infection • Caused by microorganisms already present in the patients own body • Example infection caused by intestinal flora following bowel surgery --- **4 Surgical Site Infection SSI Classification** • Superficial incisional SSI • Involves skin and subcutaneous tissue • Deep incisional SSI • Involves deep soft tissues such as fascia and muscle • Organ/space SSI • Involves any part of the body deeper than the fascia/muscle that was opened or manipulated during surgery cdc gov --- **5 According to Presence of Pus** • Non suppurative infection • Does not primarily produce pus • Example cellulitis • Suppurative infection • Characterized by pus formation • Examples • Abscess • Empyema • Infected wound

Anesthesia Types General Spinal and Local Systemic Effects 1 Definition • Anesthesia is a medically induced temporary and usually reversible state in which sensation especially pain is reduced or abolished with or without loss of consciousness to permit surgical or diagnostic procedures --- **2 Types of Anesthesia A General Anesthesia Definition** • General anesthesia is a drug induced state of unconsciousness with loss of sensation and reduced protective reflexes during surgery **Features** • Patient is unconscious • Pain sensation is suppressed • Protective airway reflexes are reduced • Muscle relaxation may be produced • Airway support/ventilation may be required **Uses** • Major abdominal surgery • Thoracic surgery • Neurosurgery • Major orthopaedic surgery • Procedures that require prolonged immobility --- **B Spinal Anesthesia Definition** • Spinal anesthesia is a regional anesthetic technique in which a local anesthetic is injected into the subarachnoid/intrathecal space producing temporary sensory and motor blockade **Features** • Patient is usually conscious • Produces numbness and weakness below the level of the block • Commonly used for lower abdominal pelvic and lower limb surgery **Common effects/complications** • Hypotension • Bradycardia • Urinary retention • Post dural puncture headache • Temporary motor weakness • Rare neurological complications --- **C Local Anesthesia Definition** • Local anesthesia produces temporary loss of sensation in a specific limited area without causing loss of consciousness **Examples** • Local infiltration for wound suturing • Dental anesthesia • Minor skin procedures • Peripheral nerve blocks **Mechanism** • Local anesthetics block voltage gated sodium channels preventing nerve impulse transmission --- **3 Systemic Effects of Anesthesia A Respiratory System** • Respiratory depression particularly with general anesthesia and sedative/opioid drugs • Reduced cough and airway protective reflexes • Hypoventilation • Reduced functional respiratory capacity • Risk of atelectasis and postoperative pulmonary complications **B Cardiovascular System** • Hypotension • Bradycardia or tachycardia • Reduced vascular tone • Changes in cardiac output • Significant spinal blockade may produce hypotension and bradycardia **C Central Nervous System** • Loss of consciousness with general anesthesia • Reduced awareness and reflexes • Drowsiness and dizziness after anesthesia • Temporary confusion or memory impairment may occur particularly in susceptible patients **D Gastrointestinal System** • Nausea and vomiting • Reduced gastrointestinal motility • Postoperative ileus may occur after major surgery **E Urinary System** • Urinary retention particularly following neuraxial anesthesia and some surgical procedures **F Musculoskeletal System** • Muscle relaxation during general anesthesia • Temporary weakness after regional/spinal anesthesia • Reduced mobility following surgery • Increased risk of deconditioning with prolonged immobilization **G Temperature Regulation** • Hypothermia may occur during and after surgery • Shivering may occur particularly after spinal anesthesia --- **4 Physiotherapy Implications** • Assess HR BP RR and SpO2 before treatment • Assess level of consciousness and neurological status • Check motor and sensory recovery after spinal anesthesia • Begin mobilization only when the patient is medically stable and adequate motor/sensory function has returned • Perform deep breathing exercises when indicated • Encourage ankle pump and limb exercises • Progress from bed mobility sitting standing walking • Monitor for • Dizziness • Hypotension • Desaturation • Excessive weakness • Altered consciousness • Use appropriate positioning and airway clearance techniques when clinically indicated /// --- **1 CELLULITIS Definition** • Cellulitis is an acute bacterial infection of the skin and subcutaneous tissues **Causes / Risk Factors** • Bacterial entry through cuts wounds or abrasions • Insect bites • Fungal skin infection • Diabetes mellitus • Lymphoedema • Poor circulation • Immunosuppression **Clinical Features** • Redness • Warmth • Swelling • Pain and tenderness • Fever and chills • Malaise • Regional lymph node swelling **Complications** • Abscess formation • Lymphangitis • Osteomyelitis • Sepsis • Necrotizing soft tissue infection • Recurrent cellulitis **Diagnosis** • Clinical examination • Blood tests if systemic infection is suspected • Wound culture if there is discharge • Imaging if deep infection is suspected **Treatment** • Appropriate antibiotics • Analgesics • Limb elevation when appropriate • Treatment of underlying conditions • IV antibiotics/hospitalization for severe infection **Physiotherapy Implications** • Assess pain swelling and mobility • Avoid vigorous massage over acute cellulitis • Elevate the affected limb when appropriate • Maintain mobility of unaffected areas • Gradually resume exercises after infection improves • Educate regarding skin care and prevention --- **2 ABSCESS 10 Marks Definition** • An abscess is a localized collection of pus within a tissue or body cavity usually caused by infection **Causes** • Bacterial infection • Infected wound • Blocked skin glands/hair follicles • Infected cyst • Foreign body • Diabetes and reduced immunity **Types** • Cutaneous abscess • Subcutaneous abscess • Deep abscess • Intra abdominal abscess • Pelvic abscess • Liver abscess • Lung abscess **Clinical Features** • Painful swelling • Redness • Warmth • Tenderness • Fluctuant swelling • Pus discharge • Fever • Malaise **Complications** • Cellulitis • Spread of infection • Sepsis • Tissue destruction • Recurrence • Fistula formation **Diagnosis** • Clinical examination • Ultrasound for selected collections • CT/MRI for deep abscesses • Culture of pus when appropriate **Treatment** • Incision and drainage is the main treatment for many abscesses • Antibiotics when clinically indicated • Analgesics • Wound dressing • Treatment of underlying disease **Physiotherapy Implications** • Avoid direct pressure over acute abscess • Avoid vigorous massage over infected tissue • Maintain mobility of unaffected areas • Provide postoperative respiratory and mobility exercises when indicated • Monitor for systemic infection • Follow wound care precautions --- **3 GANGRENE 10 Marks Definition** • Gangrene is the death of body tissue usually due to loss of blood supply infection or both **Causes** • Peripheral arterial disease • Diabetes mellitus • Severe infection • Trauma •

Atherosclerosis • Arterial obstruction • Burns or frostbite **Types** • Dry gangrene Usually due to severe ischemia • Wet gangrene Necrotic tissue associated with infection • Gas gangrene Severe bacterial infection with gas production in tissues • Internal gangrene Affects internal organs following loss of blood supply **Clinical Features** • Black brown purple or discoloured tissue • Coldness • Loss of sensation • Pain • Swelling • Blisters or ulcers • Foul smelling discharge in infected gangrene • Fever and systemic illness **Complications** • Severe infection • Sepsis • Shock • Extensive tissue loss • Amputation • Death if untreated **Treatment** • Restore blood supply when possible • Surgical debridement • Antibiotics for infection • Vascular surgery when indicated • Amputation for irreversible tissue damage **Physiotherapy Implications** • Assess circulation and functional status • Avoid pressure or trauma to ischemic tissue • Maintain ROM of unaffected joints • After amputation • Stump care • ROM exercises • Strengthening • Balance training • Prosthetic rehabilitation --- **4 TUBERCULOSIS TB 10 Marks Definition** • Tuberculosis TB is an infectious disease caused by Mycobacterium tuberculosis most commonly affecting the lungs **Types** • Pulmonary TB Affects the lungs • Extrapulmonary TB Affects organs outside the lungs such as • Lymph nodes • Spine • Bones and joints • Kidneys • Brain/meninges • Skin **Risk Factors** • Close contact with infectious TB • HIV infection • Diabetes mellitus • Malnutrition • Smoking • Immunosuppression • Overcrowded living conditions **Clinical Features** • Persistent cough • Sputum production • Coughing blood may occur • Chest pain • Fever • Night sweats • Weight loss • Loss of appetite • Fatigue **Diagnosis** • Sputum examination • Rapid molecular diagnostic tests • Chest X ray • Culture and drug susceptibility testing • TST or IGRA for TB infection **Treatment** • Combination anti tuberculosis drugs • Common first line drugs • Rifampicin • Isoniazid • Pyrazinamide • Ethambutol • Complete the prescribed treatment course to prevent treatment failure and drug resistance **Complications** • Pulmonary fibrosis • Bronchiectasis • Hemoptysis • Respiratory impairment • Pleural disease • TB meningitis • Spinal TB • Bone and joint destruction • Disseminated/miliary TB **Physiotherapy Implications** • Assess respiratory status and exercise tolerance • Monitor HR RR and SpO2 • Breathing exercises when indicated • Gradual aerobic and functional exercise • Postural and chest mobility exercises • Energy conservation techniques • Follow airborne infection control precautions for infectious pulmonary TB /////////////// --- **Hospital Acquired Infections HAI 1 Definition** • Hospital acquired infection HAI also called healthcare associated infection is an infection that a patient acquires while receiving healthcare or during/after hospitalization which was not present or incubating at the time of admission • HAIs are often preventable and are an important cause of increased morbidity mortality and hospital stay --- **2 Common Types of Hospital Acquired Infections** • Catheter associated urinary tract infection CAUTI • Associated with urinary catheterization • Surgical site infection SSI • Infection occurring at or near the surgical site after an operation • Central line associated bloodstream infection CLABSI • Bloodstream infection associated with a central venous catheter • Hospital acquired pneumonia • Pneumonia developing during healthcare/hospitalization • Clostridioides difficile infection • May occur following disruption of normal intestinal flora particularly after antibiotic exposure • Bloodstream infections • May occur from vascular access devices or other sources --- **3 Causative Organisms** • Bacteria • Staphylococcus aureus • Escherichia coli • Klebsiella species • Pseudomonas aeruginosa • Enterococcus species • Viruses • Influenza and other respiratory viruses • Fungi • Candida species • Some healthcare associated pathogens may be antimicrobial resistant making treatment more difficult --- **4 Risk Factors Patient related factors** • Extremes of age • Poor nutritional status • Diabetes mellitus • Immunosuppression • Severe underlying illness • Reduced mobility • Prolonged hospitalization **Treatment/procedure related factors** • Surgery • Urinary catheterization • Central venous catheterization • Mechanical ventilation • Prolonged antibiotic use • Invasive procedures • Poor wound care --- **5 Modes of Transmission** • Contact transmission • Direct contact between people • Indirect transmission through contaminated hands or equipment • Droplet transmission • Respiratory droplets from coughing sneezing or talking • Airborne transmission • Small infectious particles remaining suspended in air important for diseases such as pulmonary tuberculosis • Common vehicle transmission • Contaminated food water medications or equipment • Self infection/endogenous transmission • Organisms from the patients own flora cause infection especially when normal barriers are disrupted --- **6 Clinical Manifestations** Depending on the site of infection • Fever • Chills • Increased heart rate • Increased respiratory rate • Local redness warmth and swelling • Pain or tenderness • Purulent wound discharge • Cough and sputum in respiratory infection • Dysuria and urinary frequency in UTI • Altered mental status in severe infection • Sepsis in severe cases --- **7 Prevention and Control** • Strict hand hygiene before and after patient contact • Use appropriate personal protective equipment • Maintain aseptic technique • Proper sterilization and disinfection of instruments • Safe injection practices • Appropriate wound care • Proper catheter care • Remove urinary and vascular catheters as soon as they are no longer required • Appropriate isolation precautions • Environmental cleaning • Appropriate antibiotic use and antimicrobial stewardship • Surveillance and early identification of infections --- **8 Physiotherapy Implications** • Follow standard infection control precautions during treatment • Perform hand hygiene before and after patient contact • Use appropriate PPE according to the infection and isolation precautions • Assess • Temperature • HR • RR • SpO2 • General condition /////////////// --- **Stereotactic Surgery DBS Rhizotomy Thalamotomy Neural Implants 1 Stereotactic Surgery Definition** • Stereotactic surgery is a minimally invasive neurosurgical technique that uses a three dimensional

coordinate system with CT/MRI guidance to accurately locate and treat a specific area of the brain

Overview • Uses precise anatomical targeting • May use a stereotactic frame or frameless navigation • Can be used for • Deep brain stimulation • Brain biopsy • Lesioning procedures • Stereotactic radiosurgery • Treatment of selected brain tumours and vascular lesions

Indications • Brain tumours • Arteriovenous malformations • Movement disorders • Parkinsons disease • Essential tremor • Dystonia • Epilepsy • Selected functional neurological disorders

Complications • Intracranial bleeding • Infection • Seizures • Neurological deficit • Brain edema • Headache • Injury to surrounding brain structures • Radiation related complications when radiosurgery is used ---

2 Deep Brain Stimulation DBS Definition • Deep Brain Stimulation is a treatment in which electrodes are implanted into specific areas of the brain to deliver controlled electrical impulses with a pulse generator usually implanted under the skin of the chest

Overview • DBS is a reversible neuromodulation technique • Common targets include • Subthalamic nucleus STN • Globus pallidus internus GPi • Ventral intermediate nucleus VIM of the thalamus • The electrodes are connected by wires to an implanted pulse generator

Indications • Parkinsons disease • Essential tremor • Dystonia • Treatment resistant obsessive compulsive disorder in selected patients • Selected refractory epilepsy • Other indications may be considered in specialist centres

Complications • Intracranial hemorrhage • Stroke • Infection • Seizures • Lead displacement or malfunction • Hardware failure • Headache • Confusion • Speech or balance problems • Tingling or muscle contractions due to stimulation • Mood changes ---

3 Rhizotomy Definition • Rhizotomy is a surgical procedure in which selected nerve roots or their sensory fibres are partially or completely divided/ablated to reduce abnormal nerve activity or pain

Overview • It may be performed using surgical radiofrequency or other lesioning techniques depending on the nerve and condition • The aim is to interrupt abnormal sensory or motor pathways

Indications • Severe chronic pain in selected patients • Trigeminal neuralgia in selected cases • Spasticity in selected patients particularly through selective dorsal rhizotomy • Certain refractory neurological conditions

Complications • Numbness or loss of sensation • Weakness • Neuropathic pain • Infection • Bleeding • Unintended nerve injury • Recurrence of symptoms

Physiotherapy relevance • Assess sensory and motor function • Manage weakness and altered sensation • Improve balance and gait when required • Stretching and strengthening for patients treated for spasticity ---

4 Thalamotomy Definition • Thalamotomy is a surgical procedure in which a small precisely targeted area of the thalamus is destroyed or lesioned to reduce abnormal neurological activity

Overview • It is usually performed using stereotactic guidance • Modern techniques may include radiofrequency lesioning or focused ultrasound • VIM thalamotomy targets the ventral intermediate nucleus and is primarily used for tremor

Indications • Severe essential tremor • Selected tremor dominant Parkinsons disease • Medication resistant tremor when appropriate • Selected unilateral movement disorders

Complications • Speech disturbance/dysarthria • Ataxia • Balance and gait disturbance • Weakness • Sensory disturbance • Cognitive or neurological deficits depending on the target • Hemorrhage or infection with invasive procedures ---

5 Neural Implants Definition • Neural implants are medical devices implanted in or near the nervous system to record neural activity or deliver electrical stimulation to modify nervous system function

Examples • Deep brain stimulators • Vagus nerve stimulators • Responsive neurostimulation systems • Spinal cord stimulators • Cochlear implants

Indications • Parkinsons disease • Essential tremor • Epilepsy • Chronic neuropathic pain • Hearing loss • Selected movement disorders • Selected psychiatric or neurological conditions

Complications • Infection • Bleeding • Device malfunction • Lead migration or breakage • Pain at the implantation site • Nerve or brain injury • Stimulation related adverse effects /////////////// ---

Radiological Interpretation of OA and RA

1 Osteoarthritis OA • OA is a degenerative joint disease characterized by progressive cartilage loss and changes in the subchondral bone • X ray findings

- Non uniform joint space narrowing
- Osteophytes marginal bony outgrowths
- Subchondral sclerosis
- Subchondral cysts
- Bony remodeling/deformity in advanced disease

Usually affects weight bearing joints such as the knee and hip • Often has an asymmetrical distribution

2 Rheumatoid Arthritis RA • RA is a chronic autoimmune inflammatory disease affecting the synovial joints • X ray findings

- Symmetrical joint space narrowing
- Marginal bone erosions
- Periarticular osteopenia
- Soft tissue swelling
- Subluxation and deformity in advanced disease

Commonly affects MCP PIP and wrist joints DIP joints are usually spared ///////////////